

SINUMERIK 828D Smart system

Preconfigured system for standard machine tools

The right system for a smart future

- Details

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Agenda

01 **Target machine types and SINUMERIK 828D Smart system introduction**

02 SINUMERIK 828D Smart hardware and software details

03 Feed axis solution – SINAMICS S200 and SIMOTICS 1FL2

04 Spindle solution – SINAMICS S200 Spindle and SIMOTICS 1PH3 Pro

05 Cutting test result vs competitors on target machine types

06 Typical package ordering information

07 Service



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OEM key challenges in high-volume & value-oriented standard machine tool segment

Cost & Margin Pressure of OEM

- Raw materials, labor and factory operating costs keep rising continuously.
- Downstream end-users require lower machine prices, which severely compresses OEM profit margins. Traditional high-end CNC systems are too costly for large-scale standard machine production.

Performance & Stability Dilemma

- Mass production scenarios require 24/7 stable operation and minimal unplanned downtime.
- Most low-cost basic CNC products sacrifice precision and performance; while mid-range products are over-equipped with redundant functions, raising unnecessary costs. The market lacks a solution with balanced performance and cost.

Delivery & Efficiency Demands

- Emerging sectors including AI liquid cooling, data center infrastructure, automotive and 3C electronics drive exponential growth of standard machine tools.
- OEMs are forced to shorten commissioning, production and delivery cycles to meet surging orders.

Globalization & Service Barriers

- Many local OEMs intend to expand overseas business, but lack mature global service networks and unified technical support. Fragmented CNC & drive solutions bring difficulties in cross-region maintenance and technical service.

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SINUMERIK 828D Smart system - designed for standard VMC and Turning machines

System performance that refines your expectations

- **SINUMERIK 828D Smart controller**: an extension of the proven 828D platform, delivering optimized performance.
- **SINAMICS S200 servo drive system**:
  - ✓ PROFINET communication with the controller, enabling superior motion precision.
  - ✓ Built on a unified platform with planned portfolio extensions.
- **SINAMICS S200 spindle drive system**:
  - ✓ Up to 200% overload capacity, enabling fast acceleration & deceleration, higher productivity.
  - ✓ Reduced vibration for better surface quality.

Stability built for continuous production

- **Unified & Mature Platform**: 828D Smart shares a unified hardware concept and software platform with mature 828D system, ensuring a stable and consistent technology foundation.
- **Field-Proven Components**: S200 drives and motors have been widely adopted across diverse industries, with their reliability thoroughly proven in the field.
- **Robust Supply Chain**: all system components are backed by a mature production and supply chain management system, enabling timely response to market fluctuations.

Simplicity that accelerates your business

- **Simplified commissioning**: Easy and standardized configuration and commissioning is critical when commissioning machines at scale.
- **User-friendly programming and diagnosis**: faster operator onboarding and troubleshooting during large-scale ramp-ups.
- **Maintenance-Free controller design**: No fan, no battery.
- **Streamlined ordering and inventory management**: simplified options to choose from, no complex configurations needed.

Global Strength to Empower Overseas Expansion

- Powered by Siemens' global service network and international brand influence — full support for OEM's overseas business.

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SINUMERIK 828D Smart system - target Industries
- Configuration tailored to value-oriented market needs

Focus on **standard machines** tool usage **sector**,

offers **reliable, precise and efficient CNC performance.**





Support on Turning, Milling and Grinding technology






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General machining, AI liquid cooling, humanoid robotics, New Energy Vehicle, 3C, mold& die, semiconductors and more

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SINUMERIK 828D Smart system - target machine types

Main Machine Types



VMC<1.3m



2-axis Turning

Standard



1 or 2-Channel Turning center



Grinding



Gantry machine



VMC≥1.3m



Tapping center

Midrange

All types of machines...

High-End

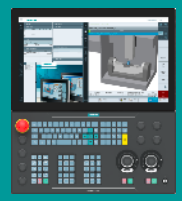
Solution



SINUMERIK 828D Smart



SINUMERIK 828D



SINUMERIK ONE

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SINUMERIK 828D Smart system - position inside SINUMERIK family

Entry class

Will phase out in Oct. 2027

SINUMERIK 808D



- Panel based
- Turning and Milling
- Up to 6 axes/spindles
- 1 channel
- 8,4"- display
- SIMATIC S7-200 based PLC

Standard

SINUMERIK 828D Smart



- Panel based
- Focus machine type: Standard vertical machining centers, Lathes for basic turning applications, Engraving
- Pre-defined system performance with selected drive and motor
- Up to 5 axes/spindles
- 1 channel
- 12,1" display
- SIMATIC S7-200 based PLC
- Cybersecurity & basic Safety function
- SW 24x

Midrange / Compact class

SINUMERIK 828D



- Panel based & Standalone
- Focus machine type: High precision machining centers, Grinding, Gantry machine, Tapping center
- With extensive drive and motor portfolio
- Up to 18 axes/spindles
- 2 channels
- 12,1" / 15,6"- multi-touch display
- SIMATIC S7-200 based PLC
- New faster CPU
- Digital Twin supported
- Cybersecurity & Safety function
- SW 24x, SW 26x, SW 28x

Premium class

SINUMERIK ONE



- Drive based
- Focus machine type: 5-axis machine, Multi technology
- With extensive drive and motor portfolio
- with NCU-Link up to 93 NC axis/spindle
- Up to 10 channels
- 12" to 24"- multi touch display modular
- SIMATIC S7-1500 based PLC
- Digital Twin supported
- Cybersecurity & Safety function

Machine performance (axis numbers, accuracy, flexibility and stability)

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SINUMERIK 828D Smart system - standardized configuration

Vertical Milling Center



SINAMICS S200 (PN 400V)
SIMOTICS 1FL2

SINUMERIK 828D Smart

SINAMICS S200 Spindle
SIMOTICS 1PH3 Pro

Lathes for basic turning



SINAMICS S200 (PN 400V)
SIMOTICS 1FL2

SINUMERIK 828D Smart

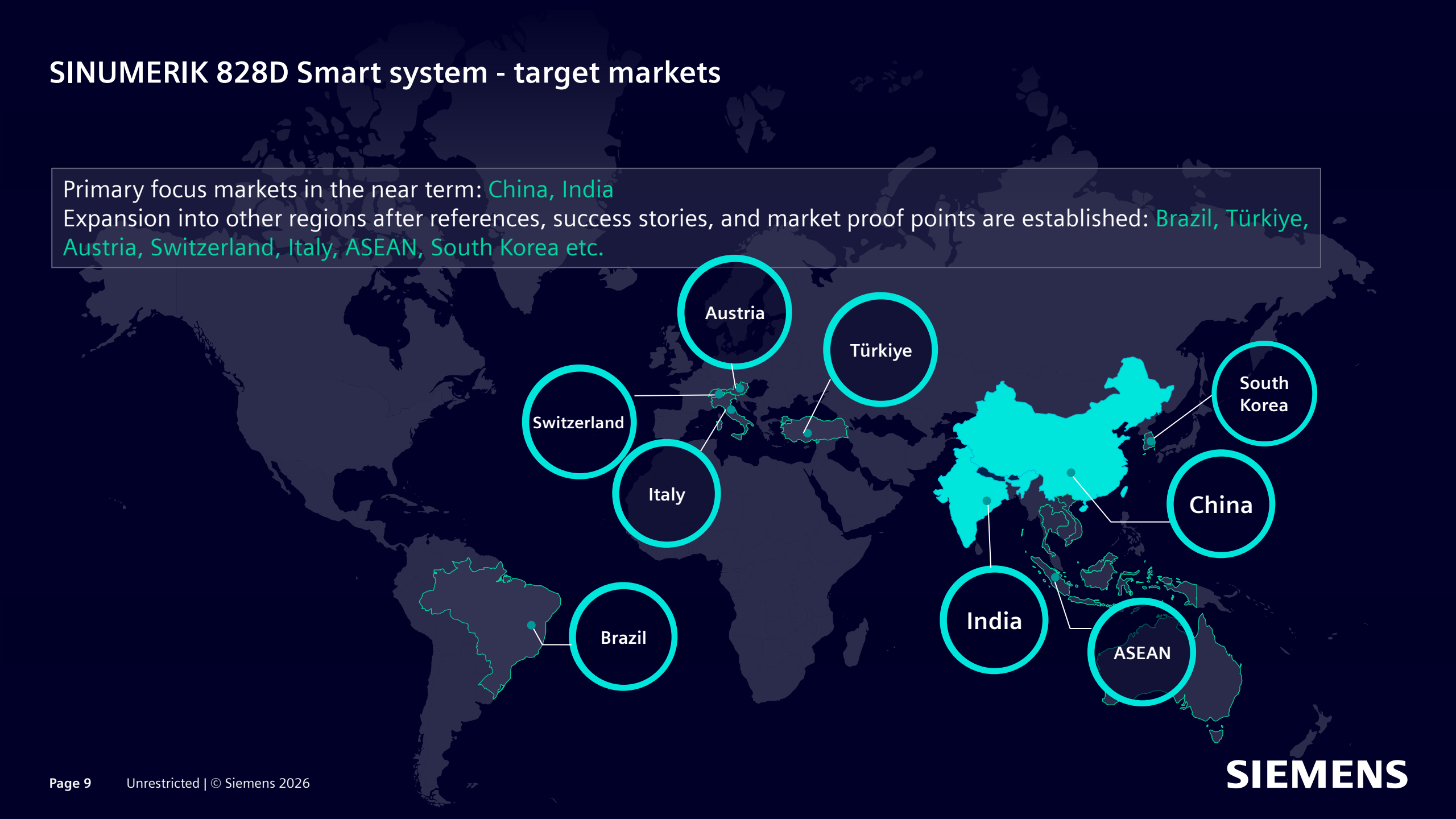
SINAMICS S200 Spindle
SIMOTICS 1PH3 Pro

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SINUMERIK 828D Smart system - target markets

Primary focus markets in the near term: China, India
Expansion into other regions after references, success stories, and market proof points are established: Brazil, Türkiye, Austria, Switzerland, Italy, ASEAN, South Korea etc.



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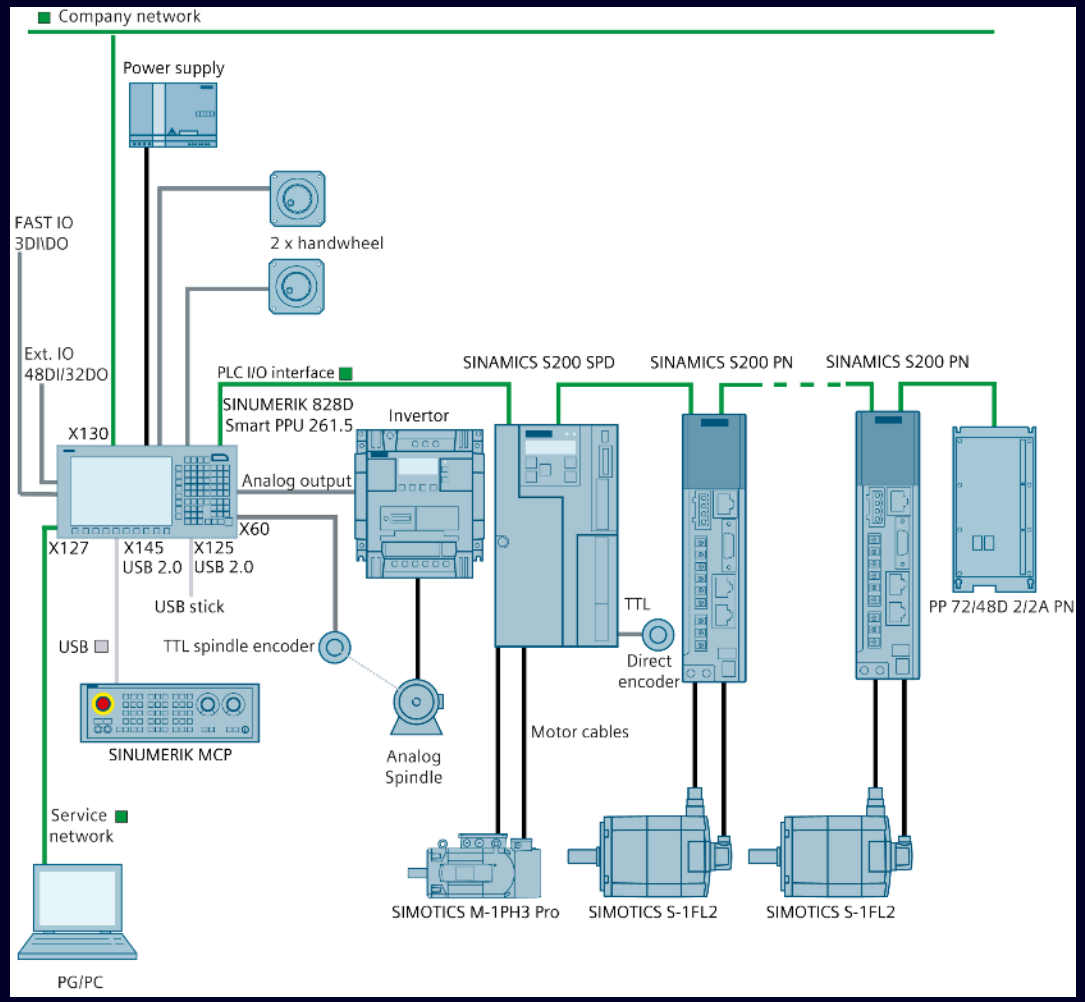
07 Service



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SINUMERIK 828D Smart PPU 26x.5 system integration
- Various connectable components



- **PROFINET connection to SINAMICS S200 & S200 Spindle**
- **SINUMERIK 828D MCP**
  - **MCP 1200U-H**
  - **MCP 1200U-V**
- **Mini handheld unit**
- **Handheld electronic handwheel**
- **PP 72/48D PN / PP 72/48D 2/2A PN I/O modules**
- **PN/PN coupler**

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SINUMERIK 828D Smart PPU 26X.5

NEW RELEASE

Standard application

Panel-based PPU



PPU 261.5



PPU 260.5

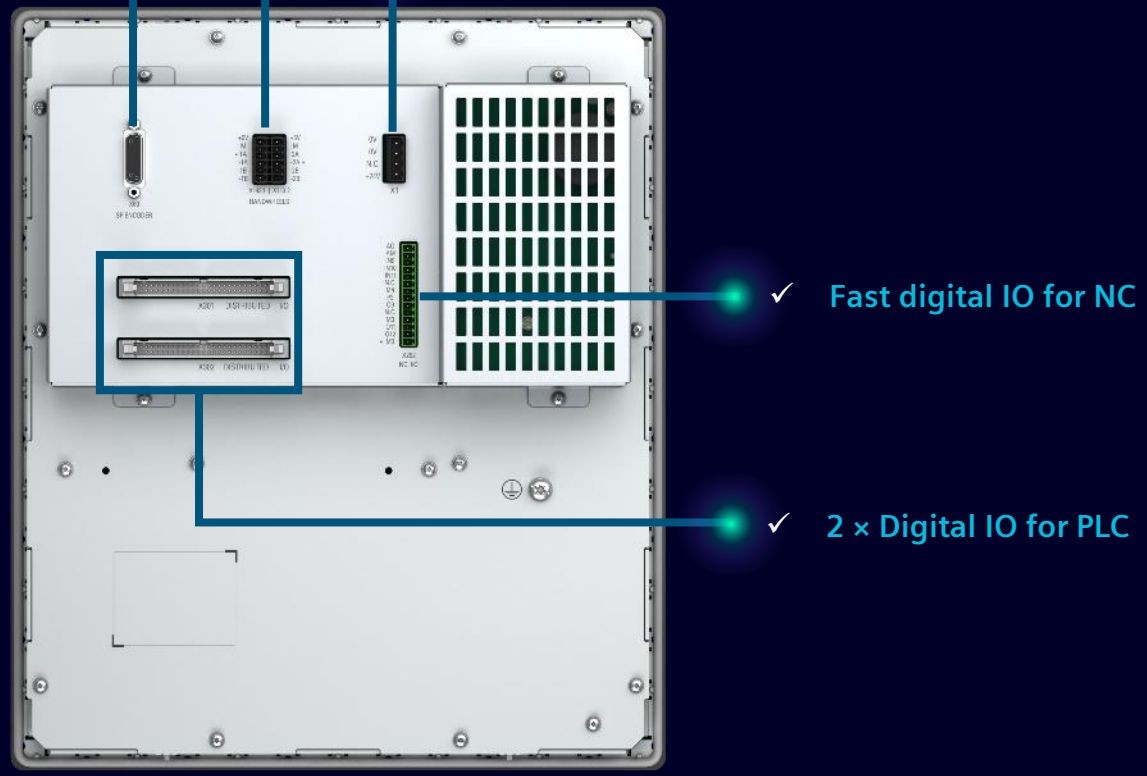
	PPU261.5	PPU260.5
CNC	16:10 1280*800 12.1" screen	
Keyboard for CNC	ONG keyboard	QWERTY Keyboard
Machine control panel	MCP 1200U-H	MCP 1200U-V
User Interface	Standard OPERATE HMI	
HMI OA tooling for Run EasyXML	Create MyScreens	
Supported software technology	SW24x T, M or G-Tec	

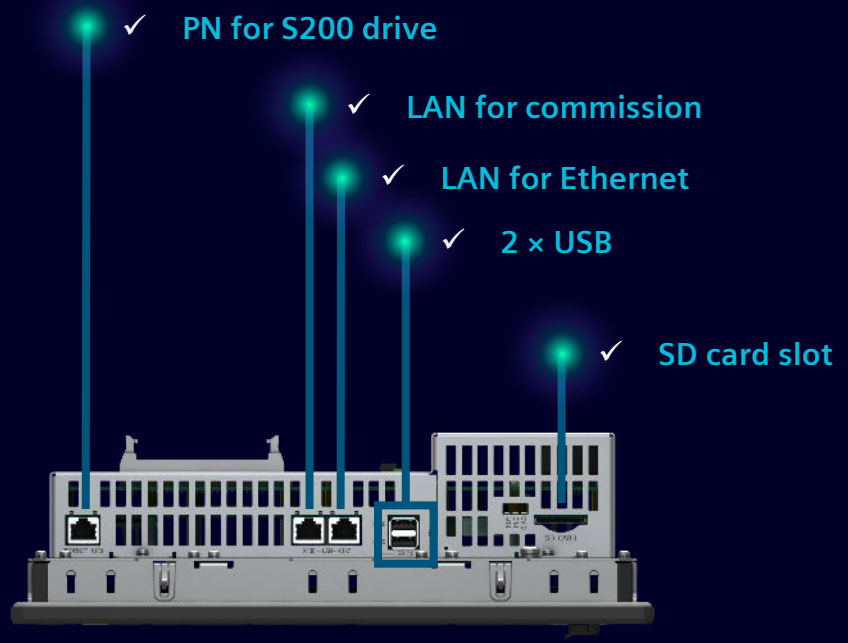
Similar design and operating style to the whole SINUMERIK 828D family

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SINUMERIK 828D Smart PPU 26X.5 system
- Interfaces just fit your demand





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SINUMERIK 828D family software scaling overview (1/3)

	Type "✓" Standard "O" Optional "– " not support	SW 24x	
		828D Smart	828D
CNC HW	PPU 260.5 / 261.5	✓	x
	PPU 270.5 / 271.5 / 290.5	x	✓
	PU 272.5	x	✓
CNC SW SW5.2x	T / M / G	✓	✓
MCP	MCP 1200 U-H	✓	✓
	MCP 1200 U-V	✓	✓
	MCP 1500U	✓	✓
Drive	SINAMICS S200/S200 Spindle	✓	–
Motor	SIMOTICS 1FL2 SIMOTICS 1PH3 Pro(Biss/TTL)	✓ ✓	– –
Operate	HT10	–	O
Axes number	Maximum axes in system (including axes/spindles in machining channel + axes in handling channel + PLC positioning axes) • Turning • Milling • G-Tech Cylindrical/G-Tech Surface	 5 5 5	 8 (5+3) 8 (5+3) 8 (5+3)

Features "✓" Standard "O" Optional "– " not support	SW 24x	
	828D Smart	828D
IPO cycle for max. configuration • Turning / Milling / G-Tech	9 ms / 9 ms / 9 ms	9 ms / 9 ms / 9 ms
Minimum block change time, approx. • Turning / Milling / G-Tech	9 ms / 9 ms / 9 ms	9 ms / 9 ms / 9 ms
Minimum block change time with compressor, approx. • Turning / Milling / G-Tech	9 ms / 3 ms / –	9 ms / 0.9 ms / –
Position control cycle	3 ms (125 µs)	3 ms (125 µs)
PLC cycle time	9 ms	9 ms
DRIVE-CLiQ S120: Velocity and current controller clock cycle	–	125 µs
Velocity and current controller clock cycle for a high-speed spindle Mixed operation without NX: 4 × 125 µs and 1 × 62.5 µs, max. number of axes = 5 • Turning / Milling / G-Tech Cylindrical/G-Tech Surface	– / – / –	– / 62.5 µs / 62.5 µs
SINAMICS S200: Velocity and current controller clock cycle for a high-speed spindle	125 µs	–
Number of Numeric Control Extensions NX10.3	–	1
Number of Numeric Control Extensions NX15.3	–	1
Number of I/O modules PP 72/48D PN or PP 72/48D 2/2A PN • Turning / Milling / G-Tech	1 / 1 / 1	3 / 3 / 3

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SINUMERIK 828D family software scaling overview (2/3)

Features "✓" Standard "O" Optional "– " not support T / M / G-Tech	SW 24x		Article No.
	828D Smart	828D	
Axis/spindle, each additional	O	O	6FC5800-0AC20-0YH0
Positioning axis/auxiliary spindle, each additional	O	O	6FC5800-0AC30-0YH0
TRANSMIT/TRACYL transformation without Y axis	O	O	6FC5800-0AS50-0YH0
Pair of synchronized axes (gantry axes), Basic	O (1)	O(1)	6FC5800-0AS51-0YH0
Generic coupling CP-Basic, e.g. multi-edge turning	O	O	6FC5800-0AM72-0YH0
Leadscrew error compensation, bidirectional	O	O	6FC5800-0AM54-0YH0
SINUMERIK Operate Runtime license OA Easy Screen Run MyScreens	O / O / ✓	O / O / ✓	6FC5800-0AP64-0YH0
Access protection for cycles Lock MyCycles	O	O	6FC5800-0AP54-0YH0
Friction compensation with adaptive characteristics	O	O	6FC5800-0AS06-0YH0
CNC lock function	O	O	6FC5800-0AP76-0YH0
Advanced technology functions (expansion of the technology cycles for turning and milling)	O / O / –	O / O / –	6FC5800-0AP58-0YH0
Extended operator functions	O	O	6FC5800-0AP16-0YH0
Basic technology functions	O	O	6FC5800-0AQ08-0YH0
Extended operator basic functions	O	O	6FC5800-0AQ06-0YH0
ShopTurn/ShopMill machining step programming	O / O / –	O / O / –	6FC5800-0AP17-0YH0
Residual material detection and machining for contour pockets and stock removal	O / O / –	O / O / –	6FC5800-0AP13-0YH0
Simultaneous recording (real-time simulation of the current machining operation)	O	O	6FC5800-0AP22-0YH0
Contour handwheel	O	O	6FC5800-0AM08-0YH0
Top Surface	– / O / –	– / O / –	6FC5800-0AS17-0YH0
Access MyMachine /OPC UA	O	O	6FC5800-0AP67-0YH0
Access MyMachine /OPC UA - Eco	O	O	6FC5800-0AQ67-0YH0

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SINUMERIK 828D family software scaling overview (3/3)

Features "✓" Standard "O" Optional "– " not support	SW 24x	
	828D Smart	828D
Tangential control Turning / Milling / G-Tech	– / – / –	– / – / O
Top Speed plus Turning / Milling / G-Tech	– / – / –	– / O / –
Nodding, Cylinder error, sag, cogging torque compensation, MEC, and spline interpolation	–	O
SMTE's and surface turning, interpolation turning, prime turning, comb grooving, turning with Y-axis, advanced rapid movement, reduced dynamic mode	–	O
Leading axis/following axis Basic for drives	–	O
Generic coupling, CP-Static, CP-Comfort	–	O
Inclined axis, inclined axis Basic fixed angle	–	O
Measuring cycle, measure kinematics	–	O
ACM, RMR/handling, PMM/3D Primitive, TCS	–	O
DXF reader, monitoring for maximum tool speed/acceleration, Path acceleration limitation, Simulation 1 (finished part) in 3D representation,	–	O
Replacement tools for tool management, Identify tool demand, Tool identification for loading/unloading tools with code carrier	–	O
ESR, IDC, ILC, Jerk adaptation, DYNNEGMA, AST addon	–	O
Angle head adapter, evaluation of internal drive variables, Ctrl-E	–	O
Memory expansion Ladder Steps, maximum (option) Basic configuration: 24000	–	100000

Features "✓" Standard "O" Optional "– " not support	SW 24x	
	828D Smart	828D
Digital inputs, maximum	120	216
Digital outputs, maximum	80	144
Analog inputs, maximum	2	6
Analog outputs, maximum (including PPU output)	3	7
Extended Safety Integrated functions: **STO, SS1, SS2, SOS, SLS, SAM, SBR, SSM, SLP, SDI, SBM, SBC, SBT**	–	O
Basic technology functions[1)]	O	✓
Extended operator basic functions[2)]	O	✓

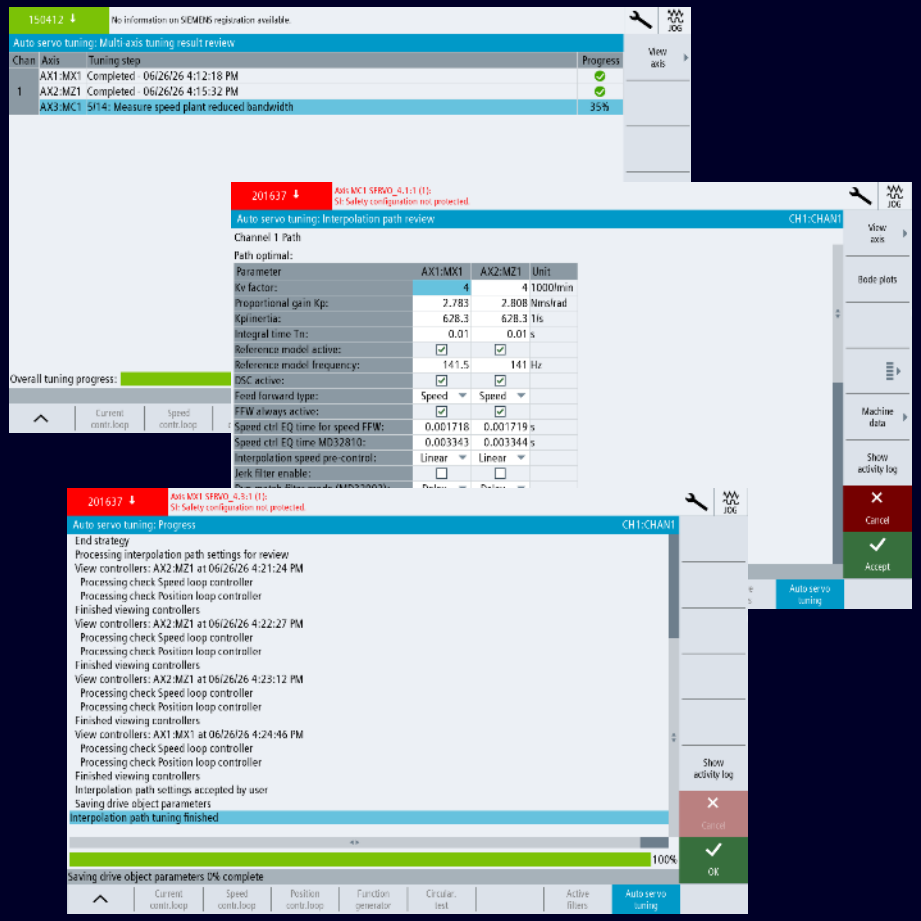
1) 6FC5800-0AQ08-0YB0: The CNC option basic technology functions provides you with technology cycles of turning and milling for the following additional machining operations:
- Swivel plane and swiveling tool
- Deep hole drilling 2
- Grid or frame position pattern
- Open slot
- Tool management function and magazine list

2) 6FC5800-0AQ06-0YB0: The operator functions in the basic scope of the SINUMERIK 828D are designed for standard applications.
The CNC option Extended operator basic functions enables the following additional operator functions:
- Positioning axes softkey in Jog mode
- Simple face milling/stock remove of the workpiece softkey in Jog mode
- Swiveling softkey in Jog mode
- Manual retraction softkey in Jog mode
- Thread synchronizing softkey in Jog mode
- Active zero offset softkey
- Overview zero offset softkey
- NC/PLC variable softkey
- Bus TCP/IP softkey
- Axis diagnosis softkey
- Additional measuring version beyond standard scope (only milling)
- Standard scope workpiece zero: Set edge, 1 circular spigot and rectangular spigot

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One-Button Auto Servo Tuning (AST)

- Delivering major time savings during commissioning for machine users



Press only one button, we handle the rest

Simply press one button; the system tunes the entire servo chain automatically within 15 minutes , bring the machine to optimal condition instantly

One-Button Start

Press AST on HMI

Intelligent Optimization

Speed, position, filters tuned

Auto Collection

Inertia, resonance

Optimum Performance

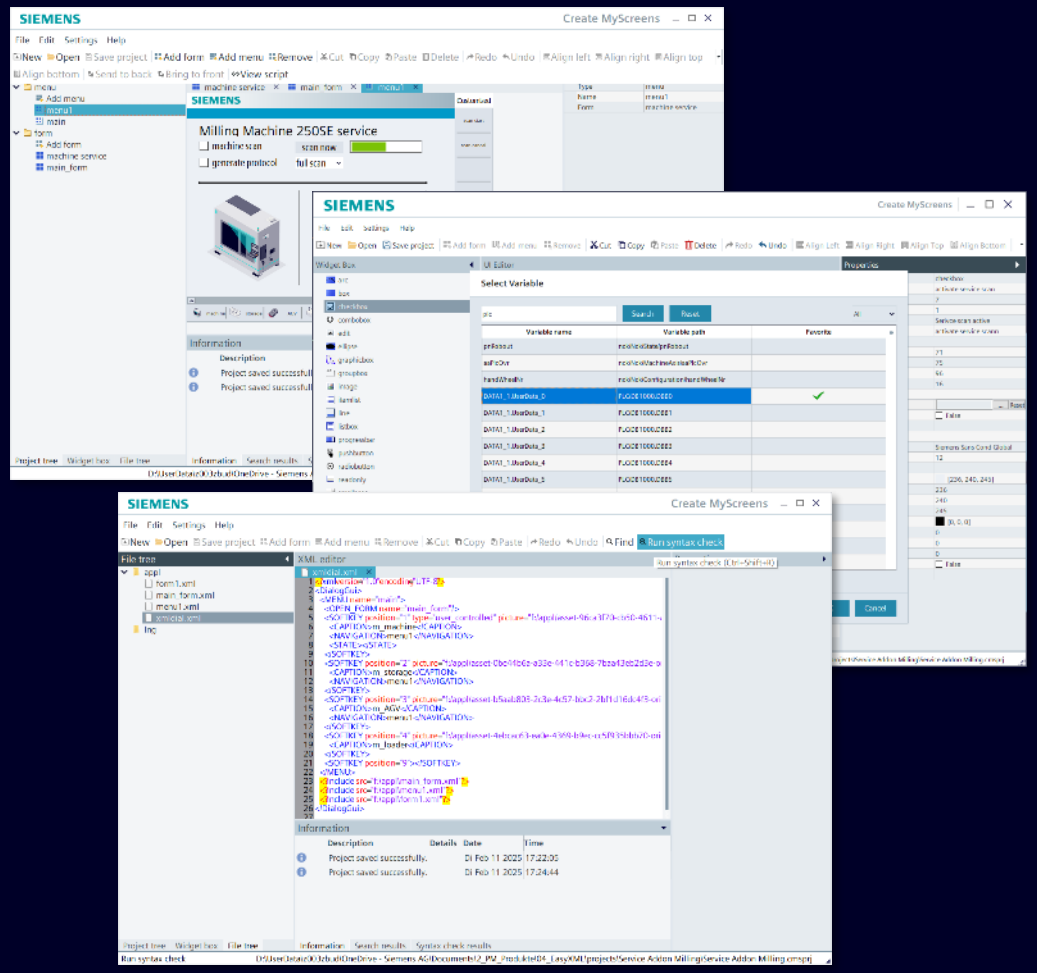
Machine at its best

- ✓ Auto position & speed tuning
- ✓ Auto filter setting
- ✓ Complete online without offline tools required

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SINUMERIK HMI: Create MyScreens (CMS) editor
- Develop your own XML screens easily



- Simplify the development task of customized screen via EasyXML
- Reduce the learning time and coding work mistakes

The editor shall enable users to design / configure screens for SINUMERIK Operate.

- Drag & Drop of elements without code writing
- Properties editor of elements
- Syntax check
- Expert editor
- ...

Create MyScreens editor is integrated in Access MyMachine /P2P software starting from version 5.0

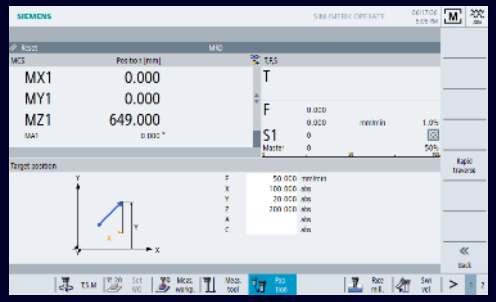
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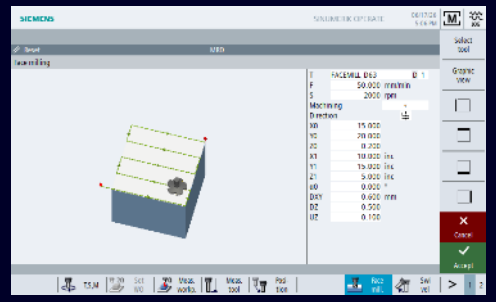
SINUMERIK 828D Smart – Extended operator basic functions
- Overview （1/2）

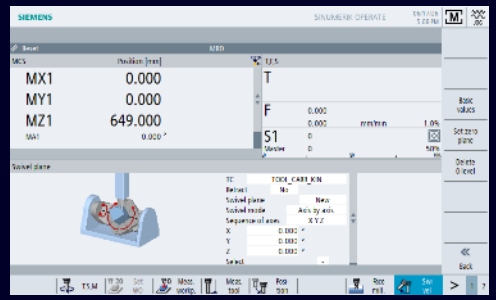
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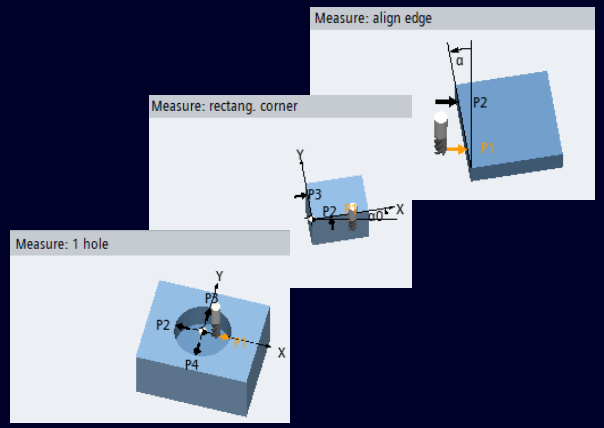
Extended operator basic functions

NEW









The CNC option Extended operator basic functions enables the following additional operator functions:

- Positioning axes softkey in Jog mode
- Simple face milling/ stock remove of the workpiece softkey in Jog mode
- Swiveling softkey in Jog mode
- Manual retraction softkey in Jog mode
- Thread synchronizing softkey in Jog mode
- Additional measuring version beyond standard scope (only milling, standard scope workpiece zero: Set edge, 1 circular spigot and rectangular spigot

- Offers a variety of easy-to-use machine setup functions for part measuring and precutting in order to reduce downtimes of the machine significantly

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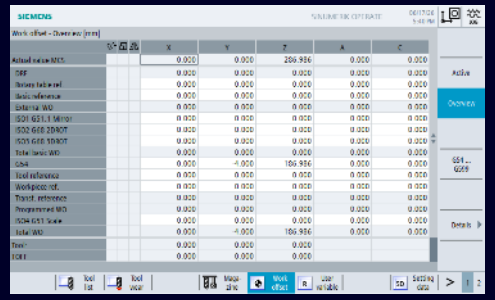
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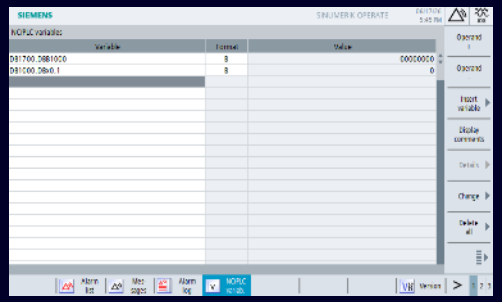
SINUMERIK 828D Smart – Extended operator basic functions
- Overview （2/2）

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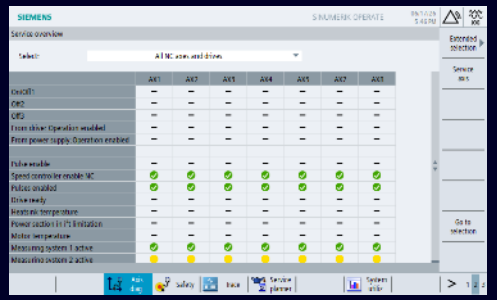
Extended operator basic functions

NEW









The CNC option Extended operator basic functions enables the following additional operator functions:

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- Active zero offset softkey
- Overview zero offset softkey
- NC/PLC variable softkey
- Bus TCP/IP softkey
- Axis diagnosis softkey

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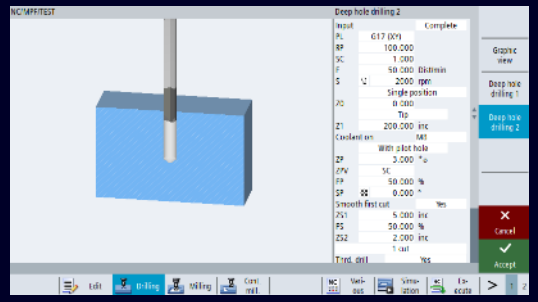
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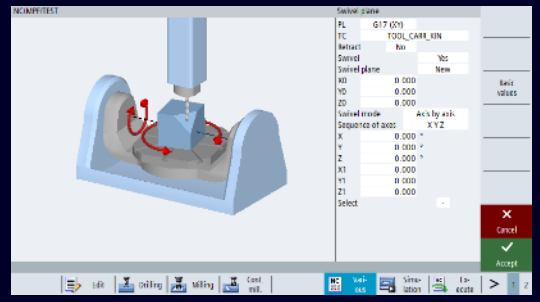
SINUMERIK 828D Smart – Basic technology functions
- Technical cycles (1/2)

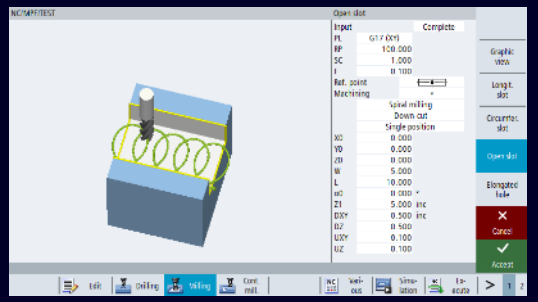
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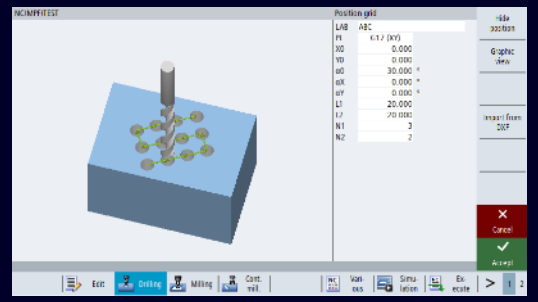
Basic technology functions

NEW









The CNC option Basic technology functions provides you with technology cycles of turning and milling for the following additional machining operations:

+

- Swivel plane and swiveling tool
- Deep hole drilling 2
- Grid or frame position pattern
- Open slot

+

- Offer more scope of machining cycles for drilling and milling of standard geometries even in swiveled planes

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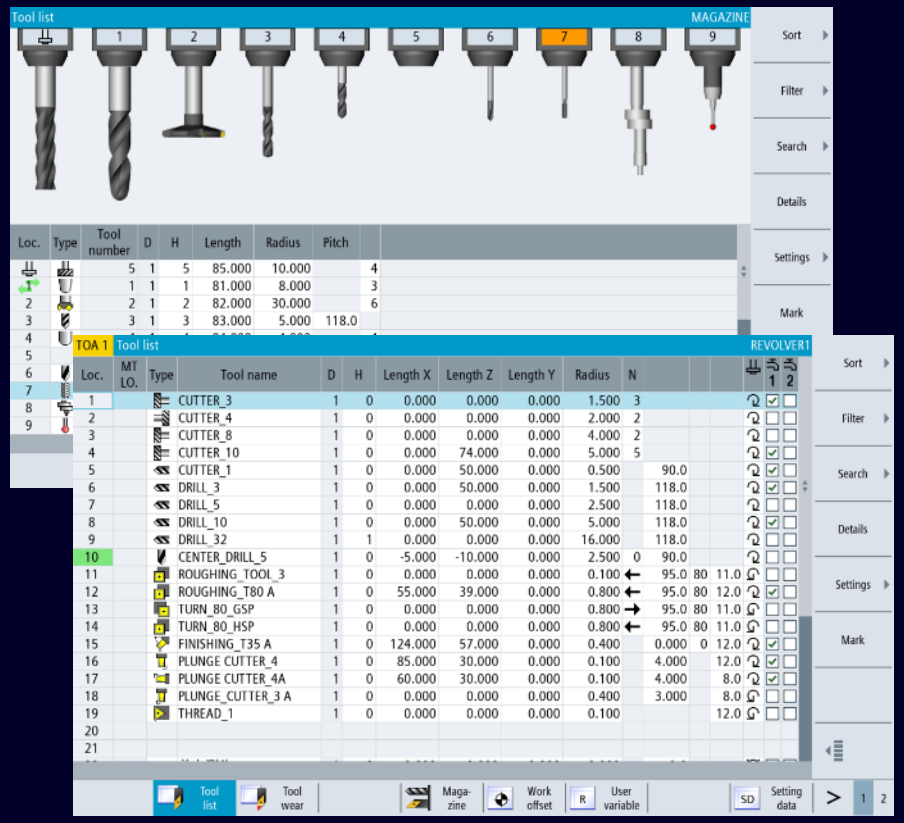
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SINUMERIK 828D Smart – Basic technology functions
- Tool management (2/2)

6FC5800-0AQ08-0YH0

Tool management

NEW



The powerful SINUMERIK tool management ensures highly productive manufacturing sequences together with user-friendly and intuitive operation :

All in one screen, all under control

- Tool types are displayed as easy to remember icons
- Clearly structured display of magazine and tool data on one screen
- Fast tool tracking due to readable tool names
- Management of up to 128 tools
- Tool life monitoring and management of replacement tools
- Convenient loading / unloading function for fast magazine assignment

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02 SINUMERIK 828D Smart hardware and software details

03 Feed axis solution – SINAMICS S200 and SIMOTICS 1FL2

04 Spindle solution – SINAMICS S200 Spindle and SIMOTICS 1PH3 Pro

05 Cutting test result vs competitors on target machine types

06 Typical package ordering information

07 Service



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SINAMICS S200 PN 400V + SIMOTICS S-1FL2
- Feed Axes



3AC 380-480V

	FSA	FSB		FSC		FSD		
$P_{rated\ [kW]}$	0.4	0.75	1	1.75	2.5	3.5	5	7
$I_{rated\ [A]}$	1.3	2.5	3.0	5.3	7.8	11	12.6	13.2
$I_{max\ [A]}$	6	9.3	11.5	20	26.5	38.1	38.8	37.4

Performance-optimized

- ✓ Speed controller cycle **125us**
- ✓ Speed up communication to controllers **with PN IRT**
- ✓ High resolution encoder up to multi-turn **21-bit +12 bit**
- ✓ Up to **350%** overload capacity
- ✓ Motors with **IP65** protection and metal connection solution

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SIMOTICS S-1FL2 high inertia(HI) servo motor
- Providing smooth running quality and low torque ripple



Shaft height SH [mm]	Power P_{rated} [kW]	Torque M_{rated} [Nm]	Speed n_{rated} / n_{max} [rpm]
45	0.4	1.27	3000 / 4000
	0.75	2.39	
65	0.75	3.58	2000 / 3000
	1.0	4.78	
	1.5	7.16	
	1.75	8.36	
	2.0	9.55	
90	2.5	11.9	2000 / 3000
	3.5	16.7	
	5.0	23.9	2000 / 2500
	7.0	33.4	2000 / 2000

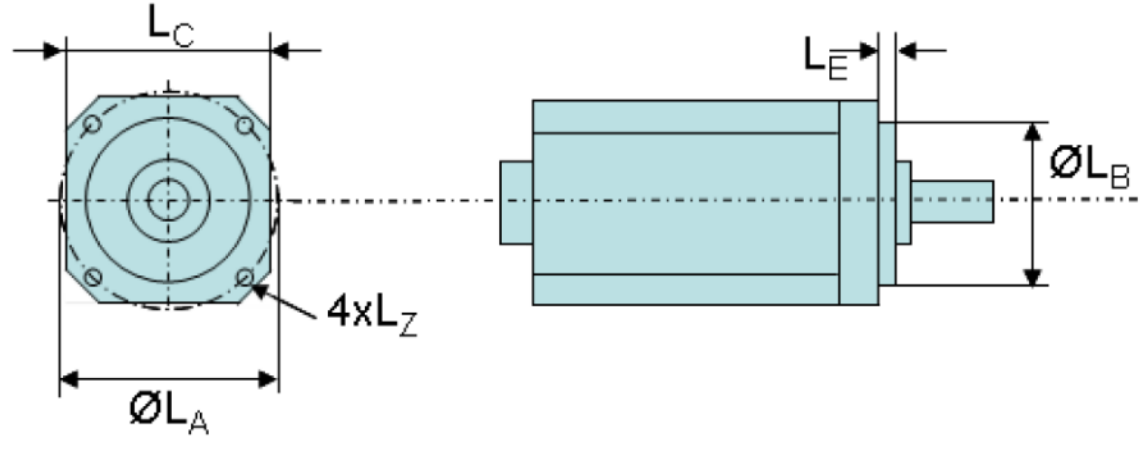
Features:

- High inertia for load adaption and smooth operation
- Max. speed up to 4000rpm
- Up to 350% overload capacity
- Protection level IP65

- Encoder
  - Absolute BiSS-C encoder multi-turn: 21-bit + 12 bit
  - Optional holding brake
- Plain shaft or shaft with key

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SIMOTICS S-1FL2 flange mechanical is compatible with other suppliers
- Also Compatible with 1FL6*



Comparison of shaft end	Descriptions		Siemens
SH65	Spigot diameter	$ΦL_B$ [mm]	110
	Spigot Tolerance	[mm]	0 / -0.035
	Spigot length	L_E [mm]	6
	Pitch diameter.	$ΦL_A$ [mm]	145
	Screw holes	$4xL_Z$ [mm]	9
	Flange thickness	LG[mm]	12

Comparison of shaft end	Descriptions		Siemens
SH45	Spigot diameter	$ΦL_B$ [mm]	80
	Spigot Tolerance	[mm]	0 / -0.03
	Spigot length	L_E [mm]	4
	Pitch diameter.	$ΦL_A$ [mm]	100
	Screw holes	$4xL_Z$ [mm]	6.7
	Flange thickness	LG[mm]	10

Comparison of shaft end	Descriptions		Siemens
SH90	Spigot diameter	$ΦL_B$ [mm]	114.3
	Spigot Tolerance	[mm]	0 / -0.035
	Spigot length	L_E [mm]	3
	Pitch diameter.	$ΦL_A$ [mm]	200
	Screw holes	$4xL_Z$ [mm]	13.5
	Flange thickness	LG[mm]	18

* 1FL6 is used in the SINUMERIK 808 system

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SINAMICS S200 and SIMOTICS S-1FL2
- Optimized feed axis solution





Wide range selections cover most applications of basic machines:

- Power range/Static torque: 0.4kW/1.27Nm ~ 7kW/33.4Nm
- Motor mechanical options: without brake / with brake; plain shaft / fitted key
- Encoder: Absolute multi-turn 21-bit + 12bit

SIMOTICS S-1FL2 Servo Motor					SINAMICS S200 Drive		
Shaft height *sh* [mm]	Power P_n[kW]	Rated Torque M_n [Nm]	Speed n_n / n_{max} [rpm]	Encoder	Frame Size	Rated Current *I* [A]	Order number
45	0.4	1.27	3000 / 4000	Abs. multi-turn 21-bit + 12bit	FSA	1.3	6SL5510-1BE10-4AF0-**Z T80**
	0.75	2.39			FSB	3.0	6SL5510-1BE10-8AF0-**Z T80**
65	0.75	3.58	2000 / 3000				6SL5510-1BE11-0AF0-**Z T80**
	1	4.78			FSC	7.8	6SL5510-1BE11-5AF0-**Z T80**
	1.5	7.16					6SL5510-1BE12-5AF0-**Z T80**
	1.75	8.4			FSD	13.2	6SL5510-1BE13-5AF0-**Z T80**
	2	9.6					6SL5510-1BE15-0AF0-**Z T80**
90	2.5	11.9					6SL5510-1BE17-0AF0-**Z T80**
	3.5	16.7					
	5	23.9	2000 / 2500				
	7	33.4	2000 / 2000				

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Agenda

01 Target machine types and SINUMERIK 828D Smart system introduction

02 SINUMERIK 828D Smart hardware and software details

03 Feed axis solution – SINAMICS S200 and SIMOTICS 1FL2

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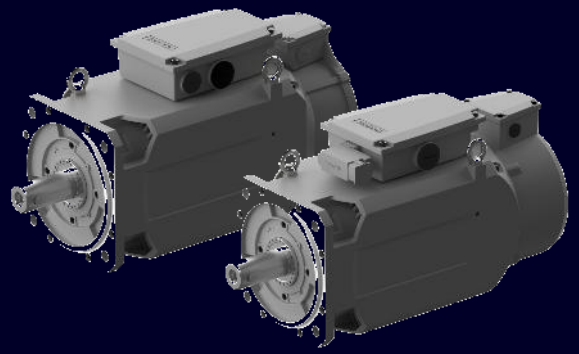
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SINAMICS S200 Spindle + SIMOTICS 1PH3 Pro
- Spindle Axis

FSE FSF



SINAMICS 200 Spindle



SIMOTICS 1PH3 Pro

3AC 380-480V

	FSE		FSF	
$P_{rated\ [kW]}$	3.7	7.5	11	15
$I_{rated\ [A]}$	13	18	30	43
$I_{max\ [A]}$	26	36	60	86

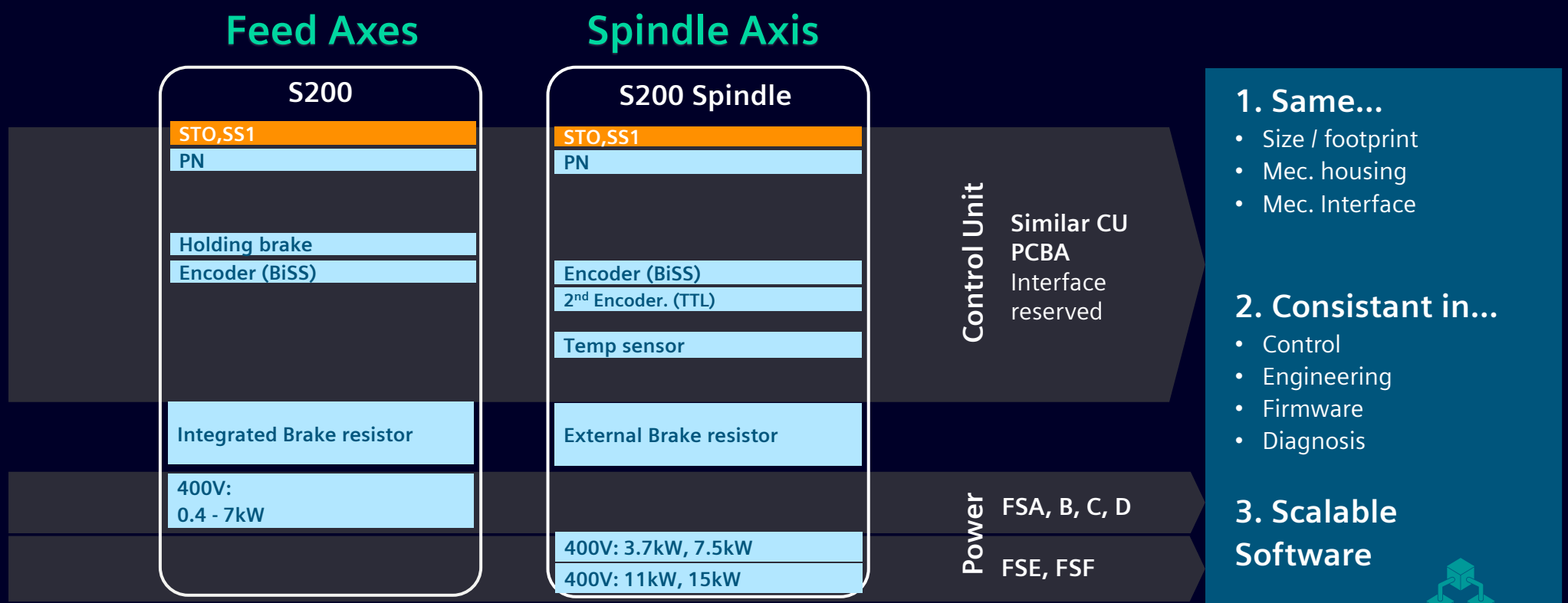
Performance-optimized

- ✓ Speed controller cycle **125us**
- ✓ Speed up communication to controllers **with PN IRT**
- ✓ High resolution encoder up to **21-bit**
- ✓ Up to **200%** overload capacity
- ✓ Motors with **IP54** protection and metal connection solution

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SINAMICS S200 family selected portfolio
- (S200 + S200 Spindle) connected with 828D Smart as M2.5 solution



- Different ...Size / footprint, Mec. Housing...
- still consistent in SW

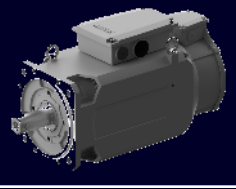
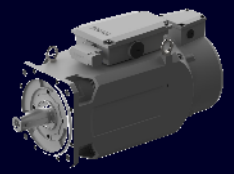
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Main features of 3rd generation SINAMICS drive S200 Spindle and S200

Topics	S200 Spindle		S200PN 400V
Supporting grid type	TN / TT / IT		TN / TT / IT
PWM frequency	4k rated	8k derating	8k
Output frequency	0~550 Hz		0~550 Hz
Current loop cycle time	125 µs	125 µs	125 µs
Speed loop cycle time	125 µs	125 µs	125 µs
Communication	PROFINET		PROFINET
Communication cycle	250 µs (IRT)		250 µs (IRT)
External 24V supply	Yes, daisy chain		Yes, daisy chain
Encoder interface	X120: BiSS X121: TTL		X120: BiSS
Commissioning interface	X127: RJ45		X127: RJ45
Motor support	1PH3 Pro		1FL2 HI
Temperature sensor	PT1000 / KTY84		/
Overload capacity	200% 2.5min		Up to 300%
Control feature	Speed control EPos Support of full closed loop control		Speed control EPos
Certification	CE, UKCA, KC, EAC, RCM		CE, UL, UKCA, KC, EAC, RCM
Ring-Redundancy	MRP, MRPD		MRP, MRPD
Safety Integrated	Basic: STO via terminal (SIL3)		Basic: STO via terminal (SIL3)
Engineering	TIA Portal/Startdrive		TIA Portal/Startdrive, Integrated Webserver
Cable Connection	up to 20m		up to 30m

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SIMOTICS M-1PH3 Pro for midrange application
- Comprehensive solution

Properties	Mid-range Main Spindle Motor 1PH3 Pro NEW!	
Applications	Entry-level series	Standard applications Turning, Grinding, entry Milling
Control system	**SINUMERIK 828D Smart**	**SINUMERIK 828D**
Converter	S200 Spindle	S120 (2nd Gen.) / S220 (3rd Gen.)
Motor		
• **Power range** • **Shaft heights** • **Variance/standards** • **Encoder** • **Accuracy** • **Vibration class** • **Overload** • **Max. speed** • **Cooling method** • **Degree of protection**	• 3.7 - 15 kW • 100/132 • no UL/STO only • TTL (Incremental) / BiSS (ST Absolute) • 13-bit, 21-bit • S, SR • 2.0 • 10,000 - 12,000 rpm • Fan cooling (NDE -> DE) • IP54	• 3.7 - 15 kW • 100/132 • UL/safety • 1Vpp Sin/Cos (Incremental) • 22-bit/DQ • S, SR • 2.0 • 10,000 - 12,000 rpm • Fan cooling (DE -> NDE) • IP55

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SIMOTICS M-1PH3 Pro main motor for SINUMERIK 828D Smart
- Flexible choice

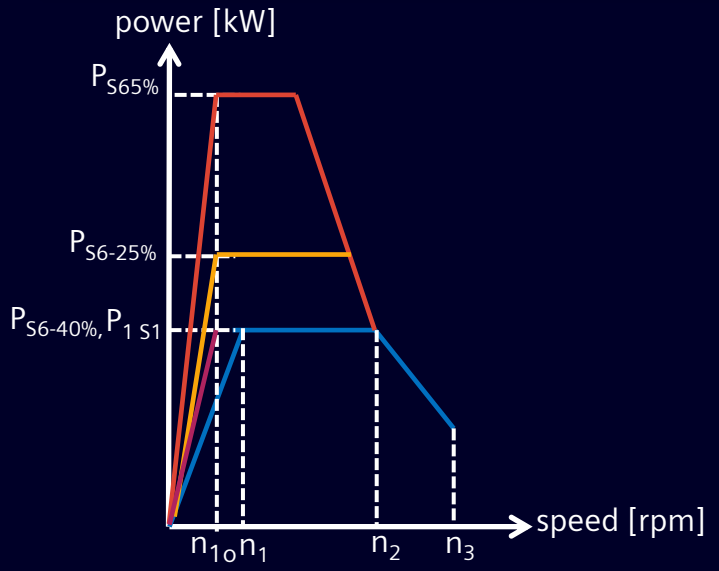
No.	1PH3..	rat. power P_{s1} / kW	rate speed n_1 /rpm	rated Torque M_{S1}	Speed n_{10}/ rpm	Speed n_2 / rpm	Max. speed n_3 /rpm	
1	101	3.7	1500	24	1500	4500	12000	
2	103	3.7	1000	35	1000	3500	10000	NEW!
3	103	5.5	1500	35	1500	4500	12000	
4	105	5.5	1000	53	1000	3000	10000	NEW!
5	105	7.5	2000	36	1500	4500	12000	
6	107	9	1500	57	1500	4500	12000	
7	107	11	2000	53	1500	5000	12000	
8	131	7.5	1200	60	1000	6000	8000	NEW!
9	132	9	1200	72	1000	5000	8000	
10	131	11	1500	70	1500	4000	10000	
11	133	11	1000	105	1000	3500	8000	NEW!
12	135	12	1000	115	1000	3400	8000	
13	133	15	2000	72	1500	4000	10000	
14	136	15	1000	143	1000	3000	8000	

- 2 shaft heights, 1 winding type, 2 encoder variants, 3 shaft variants
- New features (Balancing / Vibration / Run out / Max speed)

2 encoder variants:
- TTL: 13 bit -> 828D Smart + S200 SP (3rd Gen.)
- BiSS-C: 20 bit -> 828D Smart + S200 SP (3rd Gen.)

1 winding type:
- Y connection

3 shaft variants:
- Plain shaft
- Half-key balancing
- Full-key balancing



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SIMOTICS 1PH3 Pro with enhanced performance is competitive for standard machine Segment
- Enhanced technology

1PH3 Pro Enhanced: Run Out

- Spigot: **≤0.02mm** (1PH3 : ≤0.05mm)
- Flange axial: **≤0.02mm** (1PH3 : ≤0.05mm)

1PH3 Pro Enhanced: Rotor Quality

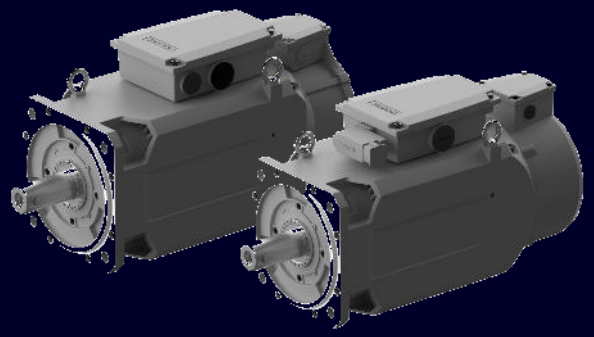
- Motor temperature **Decrease**
- Motor efficiency **Increase**
- **Enlarged** consistent power range (N2 speed)
- **Enhanced** encoder / bearing lifetime
- **Ensure** production quality consistency

1PH3 Pro Enhanced: Maximum Speed -> Standard bearing

- SH100: Up to **10krpm** (1PH3: 6krpm)
- SH132: Up to **8krpm** (1PH3: 6krpm)

1PH3 Pro

Power range	3.7kW – 15kW
Maximum speed	SH100: Up to 12krpm SH132: Up to 10krpm
Winding type	Y

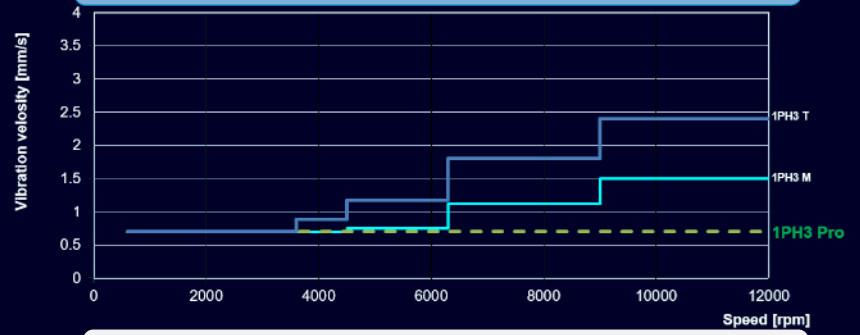


1PH3 Pro Enhanced: Packaging

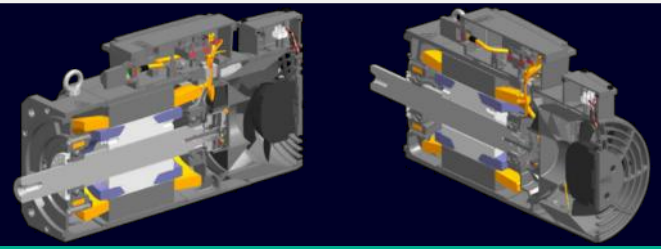


Wooden box with steel hinge / VCI bag to **Enhance** delivery protection

1PH3 Pro Enhanced: Vibration



1PH3 Pro Enhanced: On-site balance



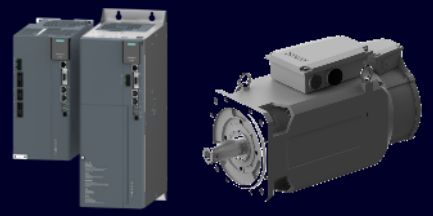
1PH3 Pro Enhanced: New encoder solution

- **BiSS-C/TTL** + S200 SP

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SINAMICS S200 Spindle and SIMOTICS 1PH3 Pro
- Optimized Spindle solution



Wide range selections cover most applications of basic machines:

- Power range with 2 different frame size: 3.7kW ~ 15kW
- Motor mechanical options: flange mounting (B5) / foot mounting (B3); plain shaft, fitted key full key balancing / fitted key half key balancing
- Encoder options: incremental encoder TTL 2500 ppr / absolute encoder single-turn 21-bit

SIMOTICS M-1PH3 Pro Main Motor					SINAMICS S200 Spindle Drive		
Shaft height SH [mm]	Power Pn[kW]	Torque (N`m)	Speed nn / nmax [rpm]	Encoder	Frame size	Rated Current I [A]	Order number
SH100	3.7	35	**1000 / 10000**	INC.2500 ppr Abs. ST. 21bit	FSE	13	6SL5510-1EE13-7AF0
	3.7	24	1500 / 12000				
			1500 / 10000				
	5.5	53	1000 / 10000			18	6SL5510-1EE17-5AF0
	5.5	35	1500 / 12000				
			1500 / 10000				
	7.5	36	2000 / 12000				
			2000 / 10000				
	9	57	1500 / 12000		FSF	30	6SL5510-1EE21-1AF0
	11	53	2000 / 12000				
SH132	7.5	60	1200 / 8000				
	9	72	1200 / 8000				
	11	105	1000 / 8000				
	11	70	1500 / 10000				
	12	115	1000 / 8000			43	6SL5510-1EE21-5AF0
	15	72	2000 / 10000				
	15	143	1000 / 8000				

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Agenda

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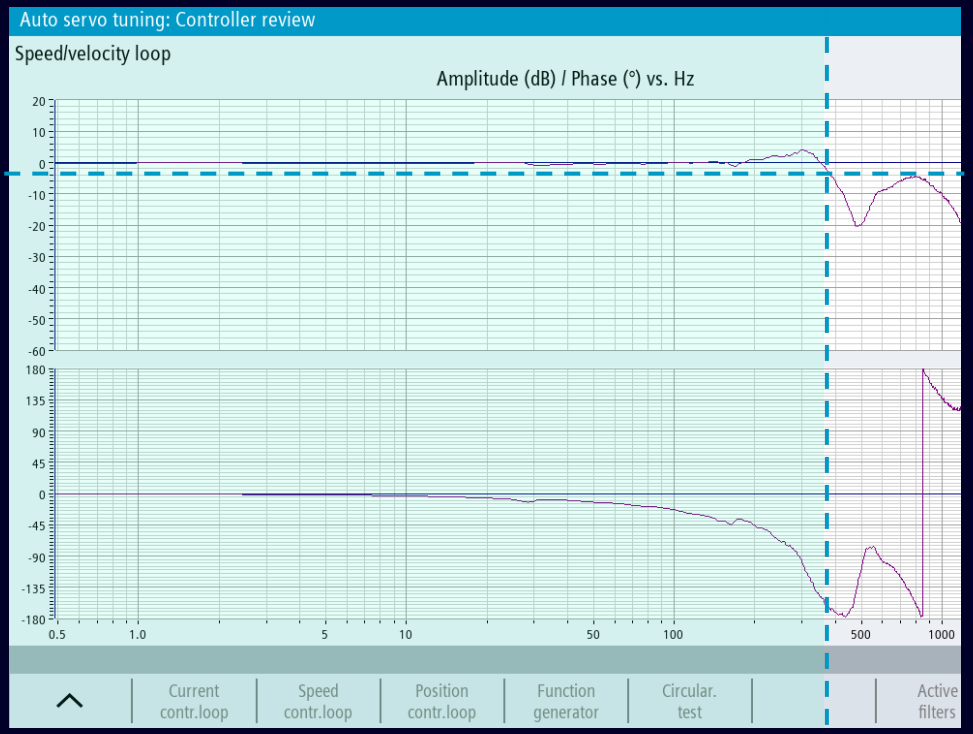
SIEMENS

High-bandwidth speed control-turning-milling composite machine
- Outstanding dynamic performance is prepared for real cutting

Turning-Milling



From AST



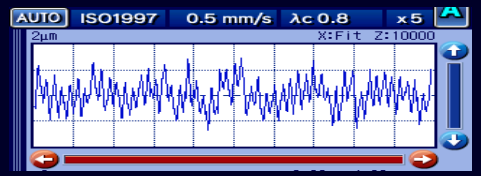
380 Hz

Bandwidth for speed control loop

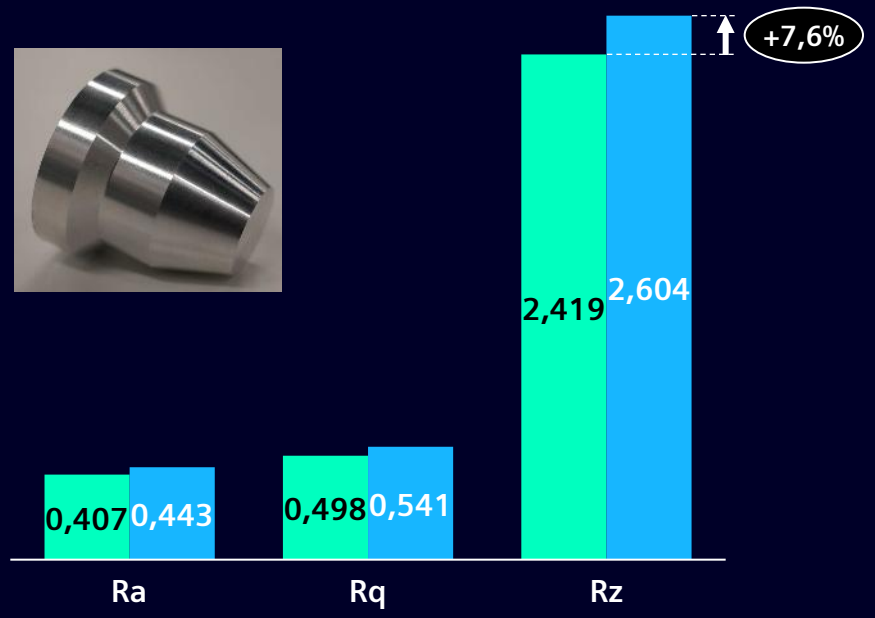
- Bandwidth frequency 380Hz
- Phase margin > 45°
- Magnitude margin > 6dB

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SIEMENS

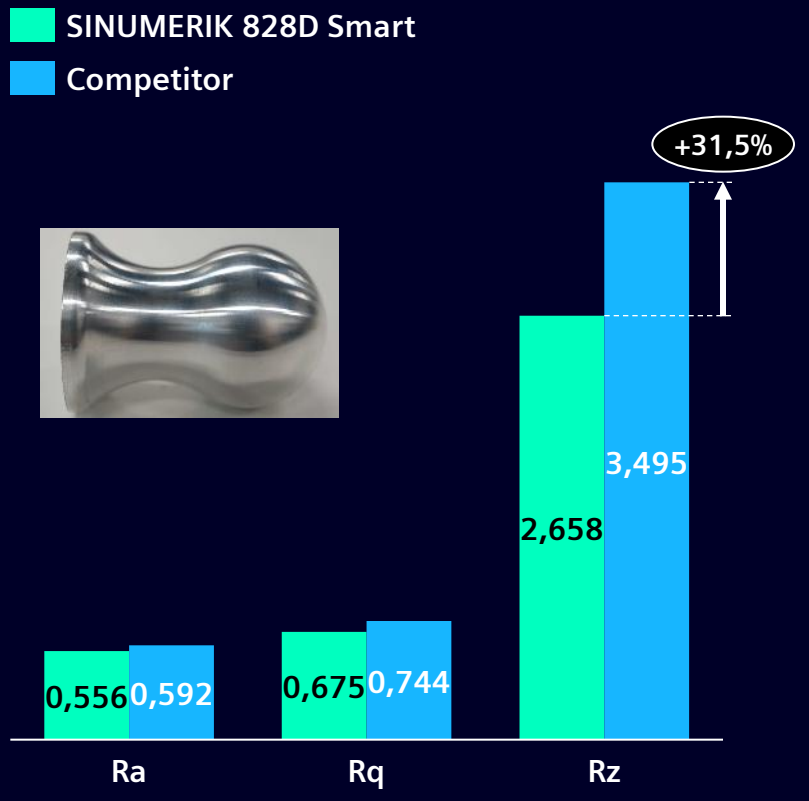
Cutting benchmark test SINUMERIK 828D Smart VS competitor in turning
- Measurement of machining result



Roughness check: µm



Siemens:**1min11s** | Competitor:**1min13s**



Siemens:**3min48s** | Competitor:**3min46s**

Key conclusion

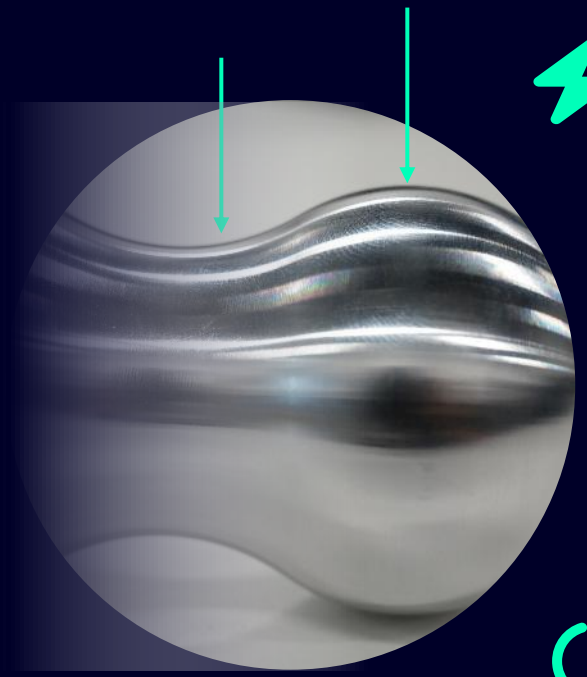
- **Visual Inspection**: Both systems are smooth and meet quality standards
- **Surface Roughness**: Siemens slightly better than Competitor — lower Rz values in both workpieces indicate better surface finish consistency
- **Machining Time**: Both systems operate at nearly the same speeds, showing the efficiency is comparable.

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The highlights of machining in turning

High Dynamic Servo Drive



High dynamic servo response
High-bandwidth speed control

Precise control of
circular interpolation

There is no friction mark
when switching quadrants

Motor pre-installed cogging torque compensation



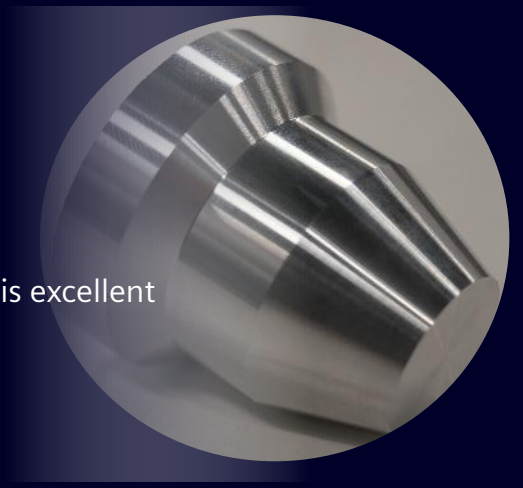
SIMOTICS 1FL2

The compensation for cogging torque before leaving the factory
Pre-calibration, install and go

The torque/speed ripple is extremely small
It operates smoothly with almost no vibration

The surface finish of conical and cylindrical is excellent
Mirror-finish surface quality

The tool path is uniform and bright
The surface is consistent and flawless

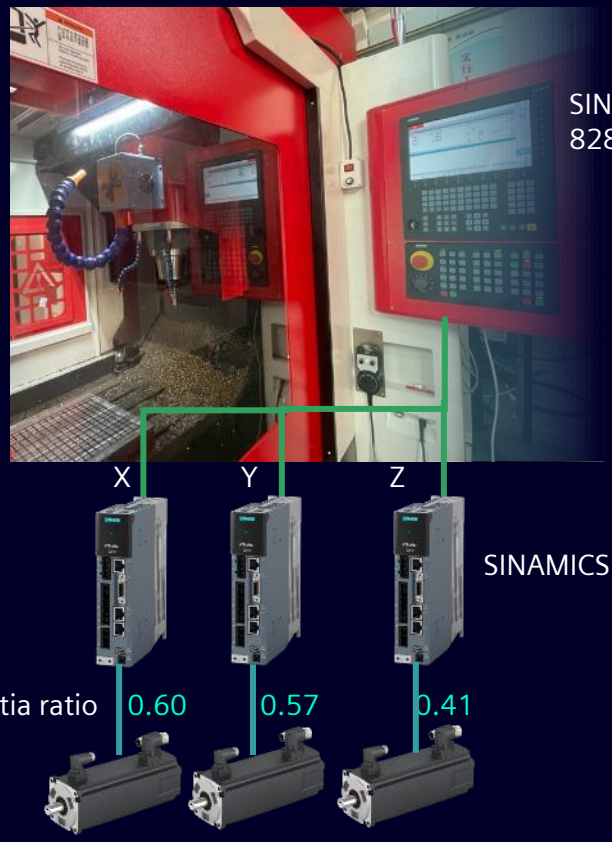


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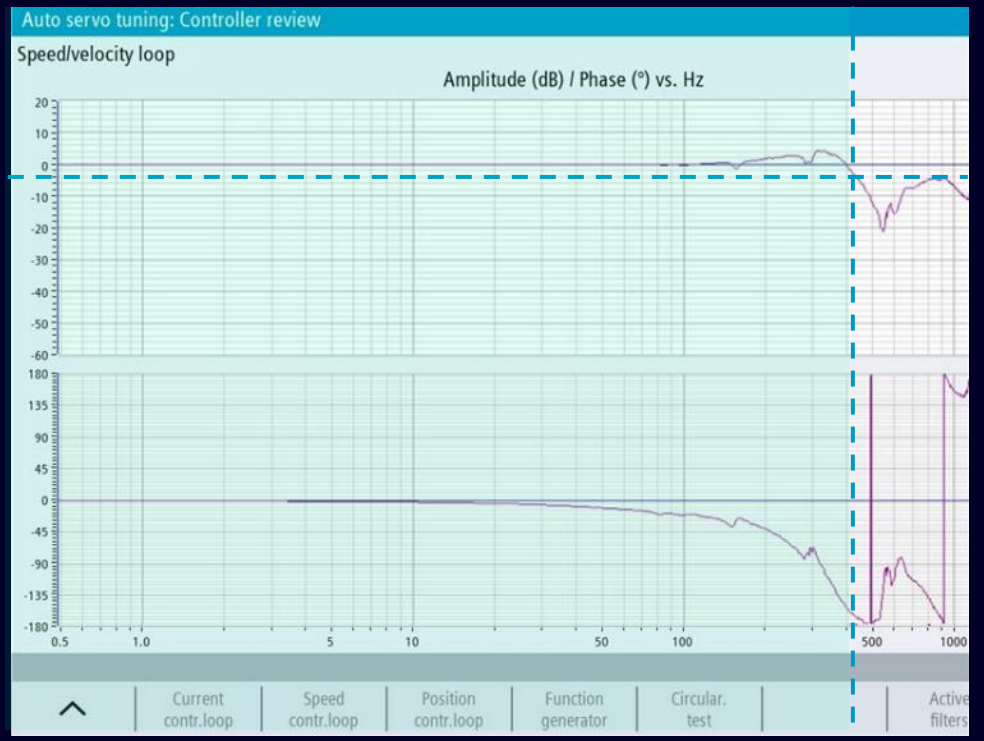
SIEMENS

High-bandwidth speed control-milling (mold-die machining)
- Outstanding dynamic performance is prepared for real cutting

Engraving Machine



From AST



420 Hz

Bandwidth for speed control loop

- Bandwidth frequency 420Hz
- Phase margin > 45°
- Magnitude margin > 6dB

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User field testing: real cutting result compared with competitor in milling



SINUMERIK 828D Smart
Advanced Surface
CTOL=0.008, 324min

The same NC program
45#Steel
S12000 F1600

The cutting path is uniform, smooth, continuous and clear.



Competitor
High Speed Setting
312 min

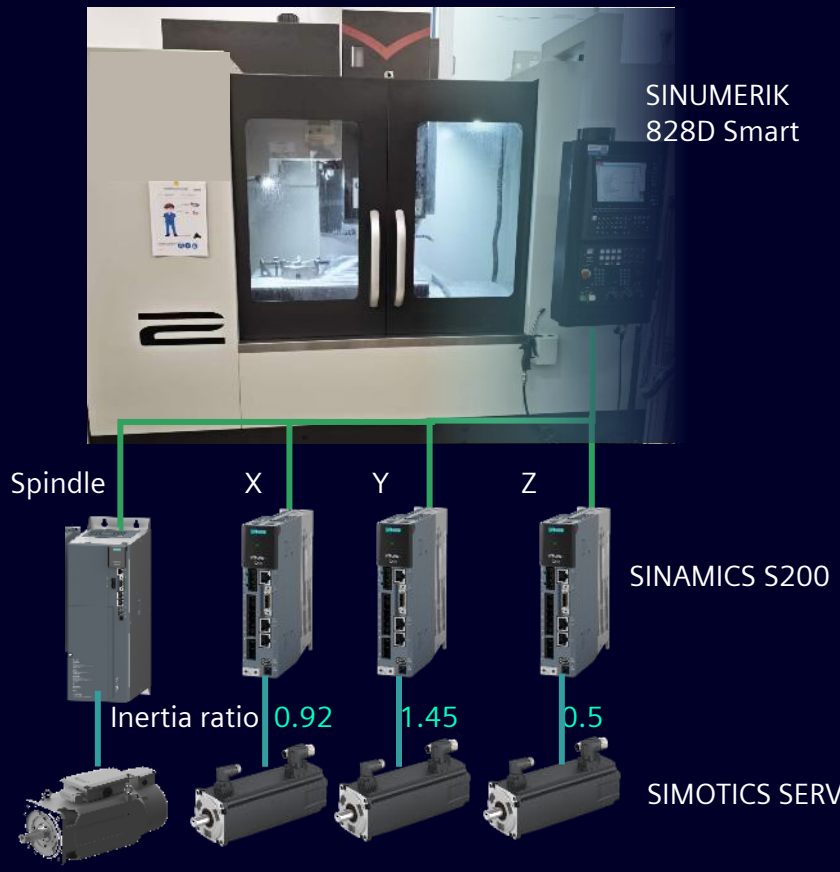
The cutting path is unclear, discontinuous, and there were over cutting

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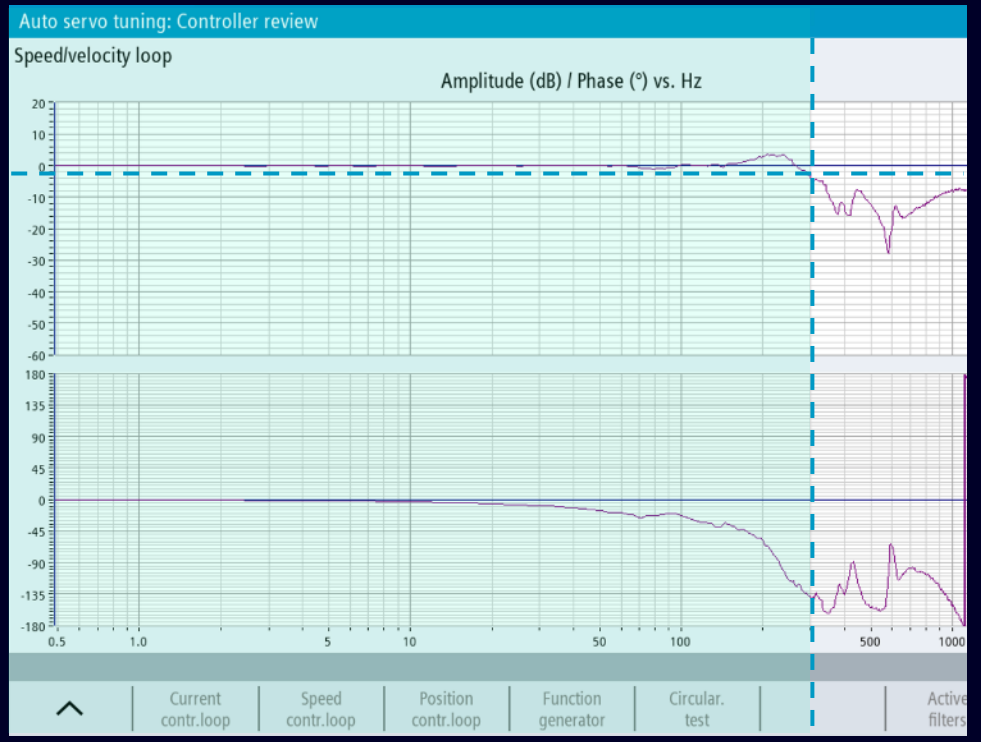
SIEMENS

High-bandwidth speed control-milling (VMC)
- Outstanding dynamic performance is prepared for real cutting

Vertical Machining Center



From AST



300 Hz

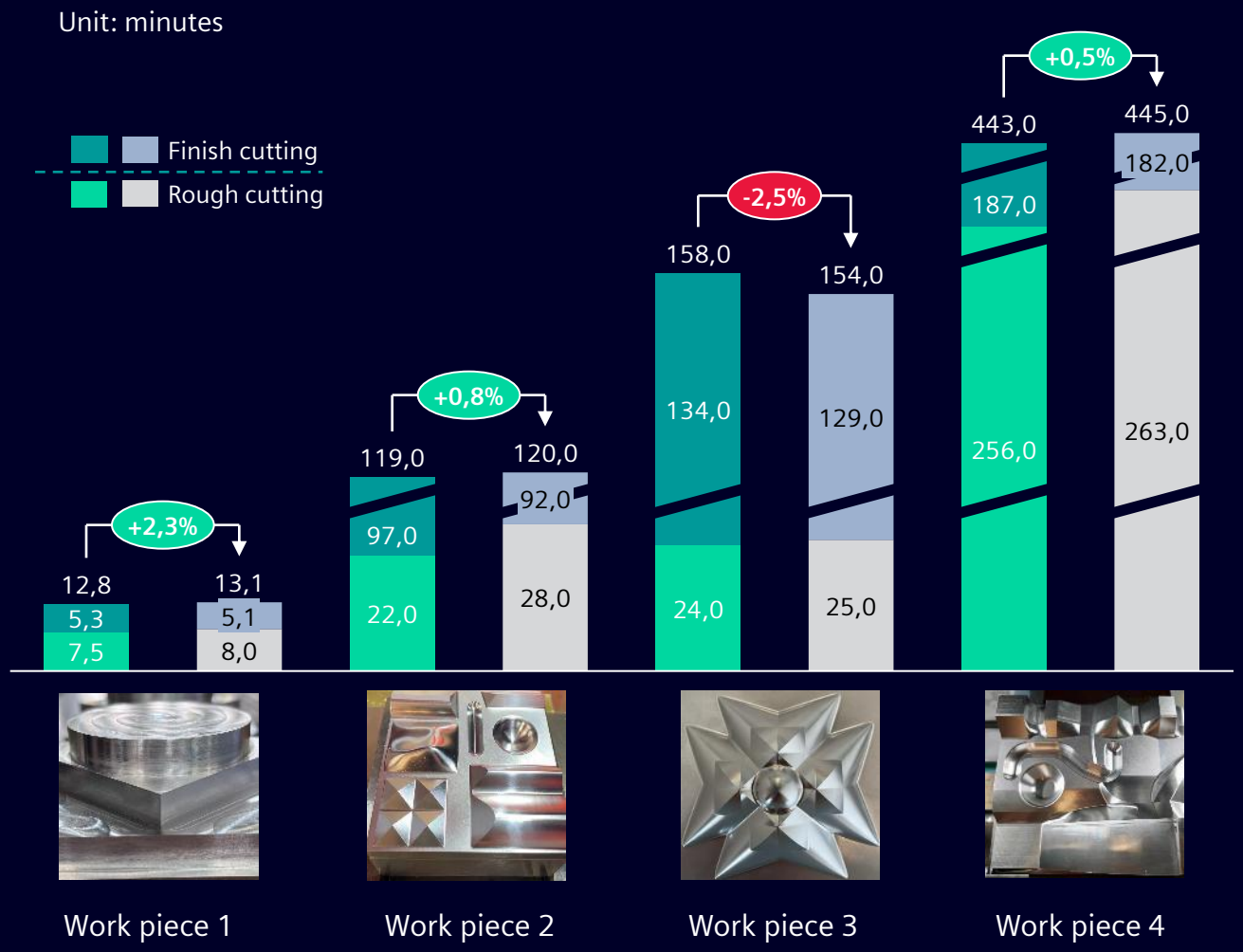
Bandwidth for speed control loop

- Bandwidth frequency 300Hz
- Phase margin > 45°
- Magnitude margin > 6dB

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Machining time performance is comparable to top competitor on typical workpiece milling manufacturing



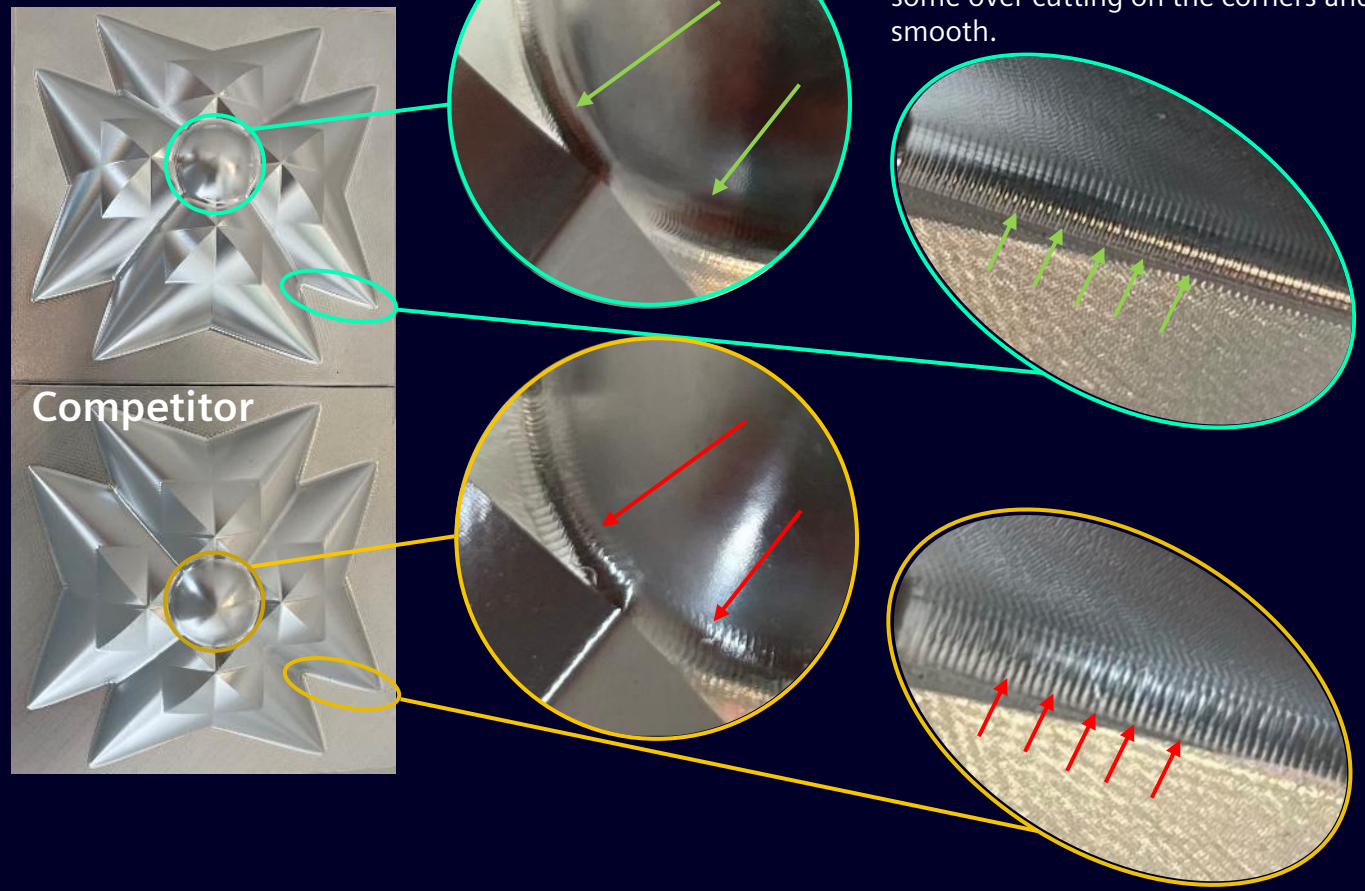


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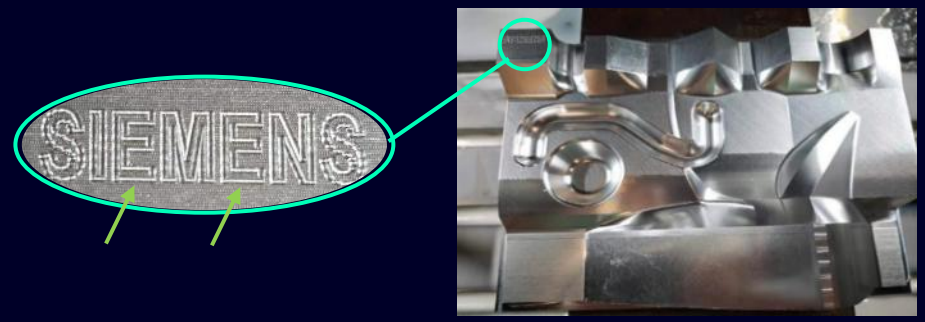
Comparison of real cutting results with competitor based on VMC855II

SINUMERIK 828D Smart
Top Surface CTOL=0.01



The cutting result for SINUMERIK 828D Smart is better in the finer details. For example, there are some over cutting on the corners and tool path is not smooth.

SINUMERIK 828D Smart
Top Surface CTOL=0.01



The cutting result for 828D smart is better in the contour precision.

Competitor



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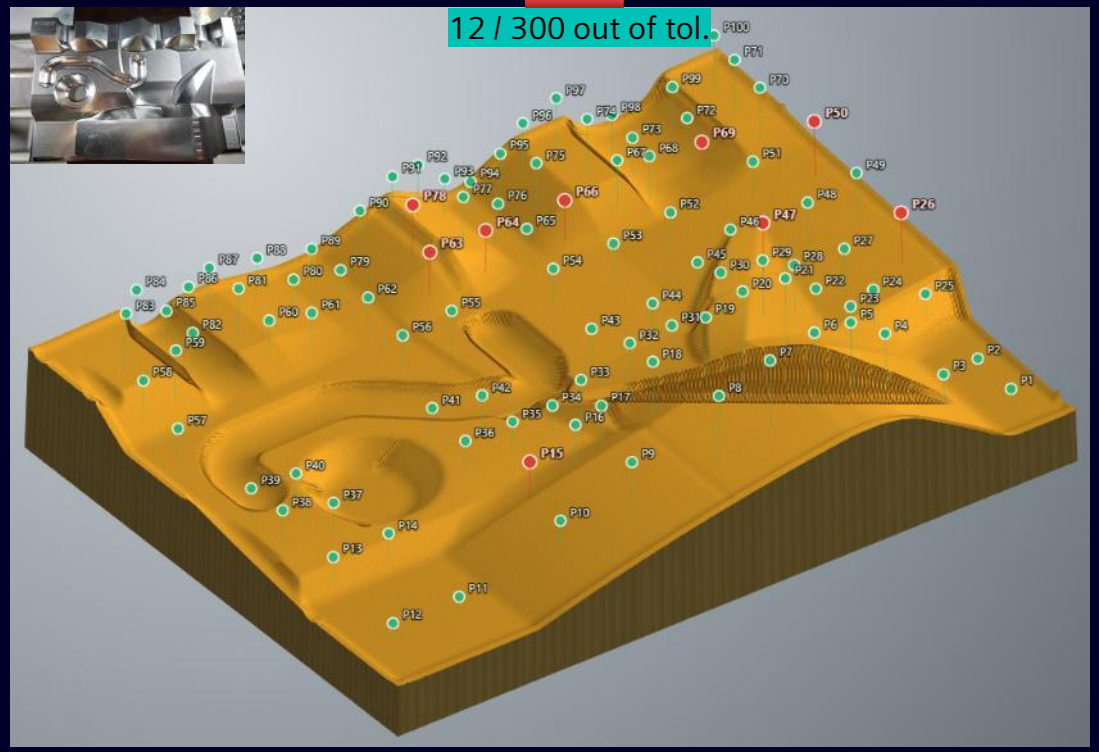
Machining accuracy comparison: Siemens vs top competitor in milling
- Red bar height = CMM out-of-tolerance ratio



300 pts· CMM

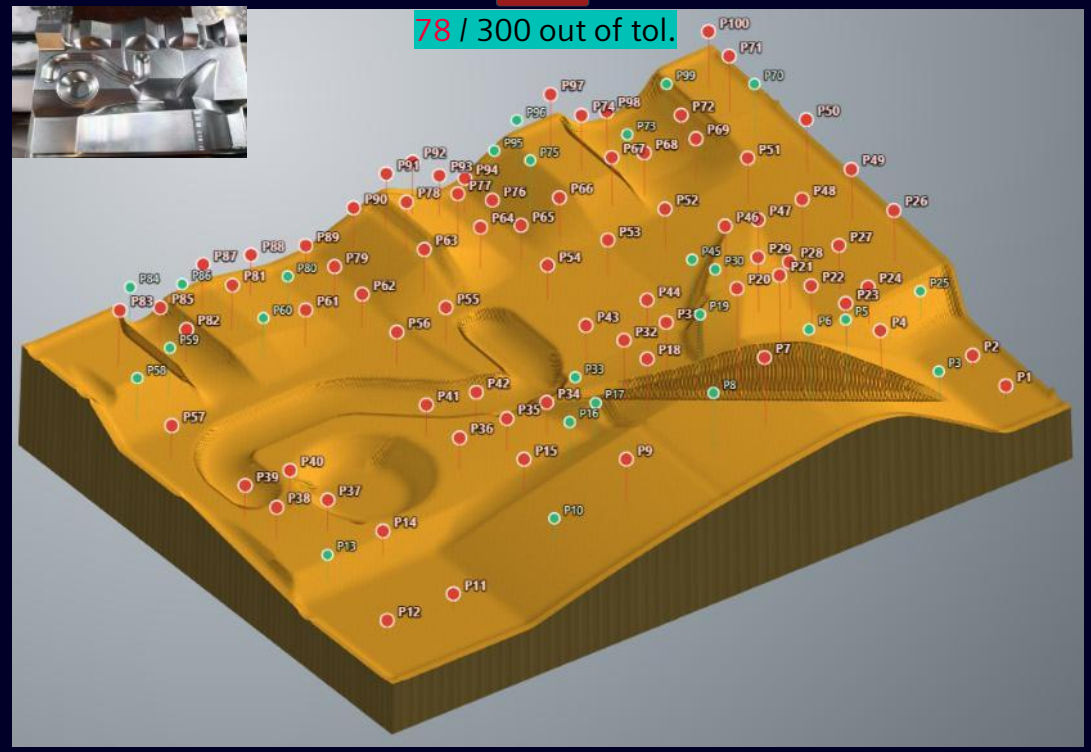
SINUMERIK 828D Smart SW24

4.0%



Top Competitor

26.0%



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01 Target machine types and SINUMERIK 828D Smart system introduction

02 SINUMERIK 828D Smart hardware and software details

03 Feed axis solution – SINAMICS S200 and SIMOTICS 1FL2

04 Spindle solution – SINAMICS S200 Spindle and SIMOTICS 1PH3 Pro

05 Cutting test result vs competitors on target machine types

06 Typical package ordering information

07 Service



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What you need in SINUMERIK 828D Smart system package

From Siemens

PPU + MCP + SD card w. SW license

(Same as PPU.5) (Same as PPU.5 SW24)



SITOP





SINAMICS S200 drive
SIMOTICS 1FL2 motor



SINAMICS S200 Spindle drive
SIMOTICS 1PH3 Pro motor

Cables
PN cable (PPU to S200)
PN cables (S200 to S200/S200 SP)
Power cables (drive to motor)
Encoder cables (drive to motor)
*Brake cables (drive to motor)

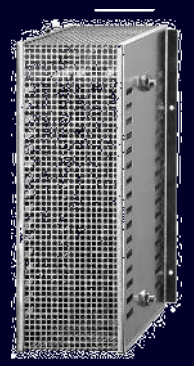


*Brake cable is only needed when 1FL2 motor is with brake.

From 3rd Party



2nd Encoder
Optional. Use a second encoder connect to PPU to achieve higher accuracy.



Brake resistor
Brake resistor **is mandatory** for S200 spindle drive!

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Typical examples for main components in a package

VMC850

Component	Description	QTY	Article No.
PPU	Vertical	1	6FC5370-3AA50-0AA0
MCP	Vertical	1	6FC5303-0AF33-1AA1
SD card with SW license	Milling	1	6FC5835-2GY50-0YA0
1FL2	M0=13.3Nm, SH65, 2kW, Nn=2000rpm, w/o. brake	2	1FL2306-7AC01-0MB0
1FL2	M0=13.3Nm, SH65, 2kW, Nn=2000rpm, w. brake	1	1FL2306-7AC11-0MB0
S200	2.5kW	3	6SL5510-1BE12-5AF0-ZT80
1PH3 Pro	SH100, 7.5kW, BISS, Nn=2000rpm, Nmax=12krpm	1	1PH3105-1HG12-0KA1
S200 SP	7.5kW	1	6SL5510-1EE17-5AF0

CK6150

Component	Description	QTY	Article No.
PPU	Horizontal	1	6FC5370-2AA50-0AA0
MCP	Horizontal	1	6FC5303-0AF32-1AA1
SD card with SW license	Turning	1	6FC5835-1GY50-0YA0
1FL2	M0=18.1Nm, SH90, 3.5kW, Nn=2000rpm, w/o. brake	1	1FL2310-2AC01-0MB0
1FL2	M0=18.1Nm, SH90, 3.5kW, Nn=2000rpm, w. brake	1	1FL2310-2AC11-0MB0
S200	3.5kW	2	6SL5510-1BE13-5AF0-ZT80
1PH3 Pro	SH132, 11kW, TTL, Nn=1500rpm, Nmax=10krpm	1	1PH3131-1LF12-0KA1
S200 SP	11kW	1	6SL5510-1EE21-1AF0

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SINUMERIK 828D Smart service concept

Warranty & service protect



- Standard: 24 months for whole system
- OSS plus: 2nd phase of 24 months
- Value added: up to 72 months

Repair service



- In-house repair as standard
- Return indicator: W55 for 828D Smart
- LRC: SNC for PPU & Drive, SMCM for motor
- RRC: After ELFF, 24 months after PM300

Technical Support



- Online support: Siemens SiePortal
- Industry Online Support Mobile app
- Siemens contact person database: Contact Database - Siemens AG

Spare & Field service



- Spare part exchange as standard, Spares on Web - Siemens AG
- Field service worldwide
- On-site support free of charge via OSS plus registration

Special Investigation



- Special Investigation as standard

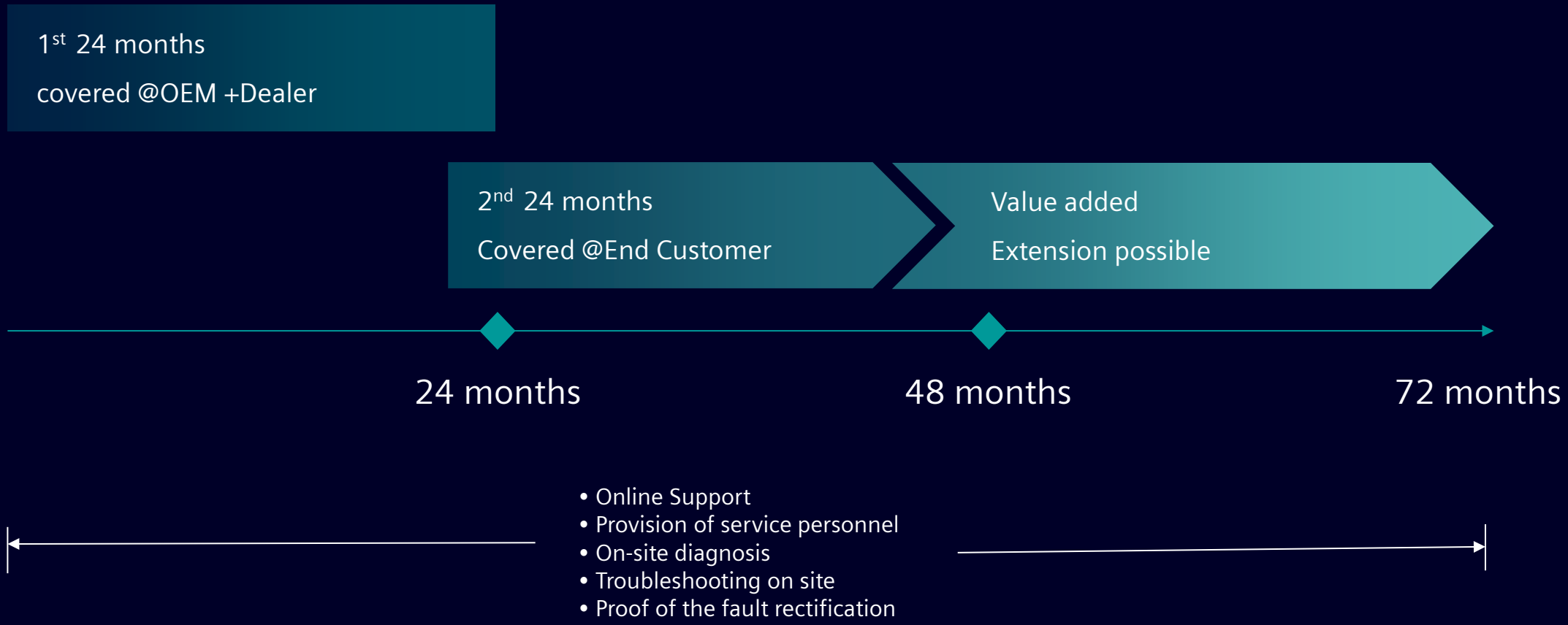
Training



- Training courses offered on SITRAIN | Siemens
- TTT for Internal training

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SINUMERIK 828D Smart warranty & OSS plus & field service



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SINUMERIK 828D Smart spare part & repair service



Spare parts

Spare part info. available:
Spares on Web - Siemens AG

Spare units:
- All module article number for whole package

Spare parts:
- Fan of 1PH3 Pro
- Fan of S200/S200 SP
- Connectors of MC350 &380

Spare kits:
- Terminal kits of S200/S200 SP
- Shield plate kits of S200/S200 SP

Repair Service & ELFF

Data regarding return or repair administration :PMD - Cockpit

ELFF:
- 24 months after PM300 for whole package

Lead repair center:
- SNC: 828D Smart & S200 & S200 Spindle
- SMCM: 1PH3 Pro & 1FL2

Regional repair center:
- India & Germany for whole package after ELFF

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SINUMERIK 828D Smart system
- Why should you use it now?



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